Sustainable Mobility: Analysing Key Factors in Choosing Electric Bikes

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Abstract

The raising impact of sustainable energy driven locomotives and the compatibility to a common man gives an essence of electric bikes. As such, these bikes gain popularity as a sustainable alternative to fuel-powered vehicles. But it is seen that there are many challenges related to the performance, breakdown, and mishappening due to technical difficulties with the electric bikes. Therefore, one needs a basic understanding to select the same to overcome the challenges from the availability. The key factors that need to monitor includes price, mileage, and performance. Hence, this study aims to develop a comprehensive idea to analyse and compare electric bikes across various parameters, including price, mileage, speed, and additional features using the proposed formula. Process of proposed formulation to obtain an optimal featured electric bike involves aggregation of data from multiple models of the bikes, objective reports tailored to user preferences and requirements. It will give a liberty to the consumers for selecting a preferred bike. This will empower users to make well-informed, data-driven decisions in selecting an electric bike with streamline the decision-making process in a rapidly growing market, enhancing customer satisfaction. Finally, the discussed approach shall provide the consumers as well as the manufacturers to make more informed and tailored decisions.

1. Introduction

Electric bikes (e-bikes) are becoming an increasingly popular, eco-friendly alternative to traditional fuel-powered vehicles (Yadav, 2020), (Nayak, 2020), (Mohammed & P.Sai Rani, 2022), (Mahapatra, A & Bansal, M, 2020), offering benefits such as lower costs, reduced pollution, and less traffic congestion. However, consumers often face challenges in choosing the right e-bike due to the variety of models available, (Chou, S & Lin, C, 2021) each with different features such as price, battery life, performance, (Christopher CHERRY & Min HE, 2009) and additional features. This lack of a clear, (Day, Nayak,

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Kumar, & Mohanty, 2024) comprehensive comparison system can lead to confusion and poor decision-making. The aim of this research is to develop a framework that helps consumers compare e-bikes effectively by analyzing key parameters such as price, (Guo, Q & Zhang, R, 2021) mileage, speed, and battery life. The study will gather data on various e-bike models, (Jain, R & Singh, S, 2022) weigh the importance of each feature based on consumer preferences, (Lee, K & Lee, Y, 2019) and create a decision support tool that allows users to easily compare e-bikes. Previous studies have shown that factors like price, performance, and reliability play a major role in e-bike adoption. (Liu, X & Zhang, Z, Optimization of Electric Bike Design for Enhanced User Experience and Performance, 2021) Challenges such as battery efficiency, motor performance, and safety concerns have also been identified as barriers to widespread use. (Zhao, X & Liu, P, 2019) Decision support systems, often used in the automotive sector, have been shown to help consumers make more informed choices, and this research aims to adapt these methods for electric bike selection. (Zhang, Y & Zhao, H, 2020) The contribution of this study will be to provide a data-driven, user-friendly method to help consumers make better purchasing decisions and to offer manufacturers insights into consumer preferences, ultimately leading to better products and higher customer satisfaction. (Zhang, J & Song, Y, 2019) (Wojciech Sałabun, Krzysztof Palczewski, & Jarosław W˛ atróbski, 2019), (Wang, W & Yu, Z, 2021), (Shen, L & Yang, Y, 2020), (Saha, A & Ghosh, R, Assessing the Feasibility and Consumer Adoption of Electric Bikes in Urban India: A Mixed- Methods Approach, 2020),

2. Literature review:

Many studies highlight the growing popularity of electric bikes (e-bikes) due to their environmental benefits, cost savings, and reduced traffic congestion. However, consumers often face challenges in choosing the right e-bike due to varying features like price, battery life, and performance. Research shows that factors such as cost, convenience, and safety strongly influence consumer decisions. Some studies have proposed decision support systems to help consumers compare e-bikes based on key criteria. This research aims to develop a simple, data-driven tool to help consumers make better choices when selecting an e-bike.

3. Model Summary

The model used is Multiple Linear Regression (MLR) with Ordinary Least Squares (OLS) estimation, implemented through the statsmodels library. MLR is chosen because it allows the analysis of the relationship between multiple independent variables (Range, Top Speed, and Charging Time) and the dependent variable (Price). This approach is ideal for predicting price based on several factors simultaneously. It estimates the effect of each independent variable on the price while controlling for the others. MLR also provides coefficients that quantify the strength and direction of each relationship. The model is suitable because it assumes a linear relationship between the variables, which is a reasonable assumption for pricing electric bikes based on these factors. It enables us to determine how changes in range, speed, and charging time affect the price. The OLS method ensures that the model minimizes the sum of squared errors to provide the best-fitting line.

4. Data Analysis and Interpretation

The data analysis for this study will focus on understanding the relationships between key features of e-bikes such as price, mileage, speed, and battery life. A heatmap will be generated to examine the correlation between these features, providing insights into how they interrelate and influence each other. For example, we might observe a high correlation between price and battery life, suggesting that more expensive e-bike models tend to have better battery performance. Additionally, regression plots will be used to explore the predictive relationships between independent variables (such as price and battery life) and dependent variables (such as mileage or speed). Scatter plots will highlight the distribution of data points, showing trends and outliers that may affect consumer decisions. These visualizations will allow us to identify patterns, such as whether higher speed correlates with lower battery efficiency or if longer battery life is associated with higher cost. The goal of this analysis is to derive actionable insights that will inform the development of the decision support tool, ensuring that it reflects the most important factors influencing e-bike purchases. By understanding these relationships, the tool will better guide consumers in making informed decisions based on their priorities.

Ather :

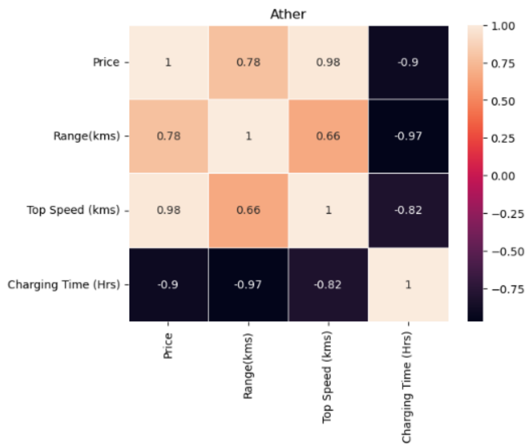

Fig. 1. Correlation plot of Ather bike models

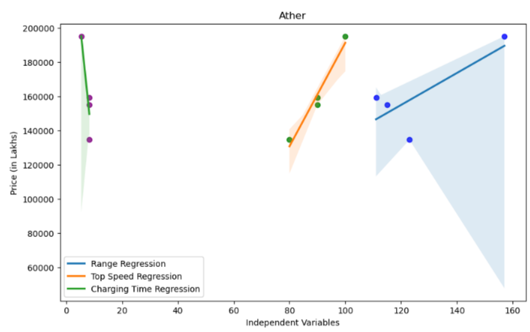

Fig. 2. Regression plot of Ather bike models

Fig. 1. The heatmap reveals a strong positive correlation between price and top speed for Ather electric bikes. Conversely, a strong negative correlation exists between charging time and both range and top speed.

Fig. 2. The plot illustrates the relationships between price and three independent variables (range, top speed, and charging time) for Ather electric bikes. The regression lines and confidence intervals suggest that price increases with range and top speed but decreases with charging time.

Hero :

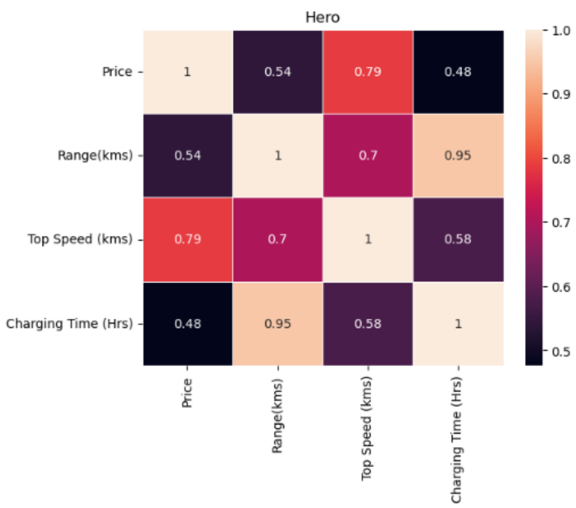

Fig. 3. Correlation plot of Hero bike models

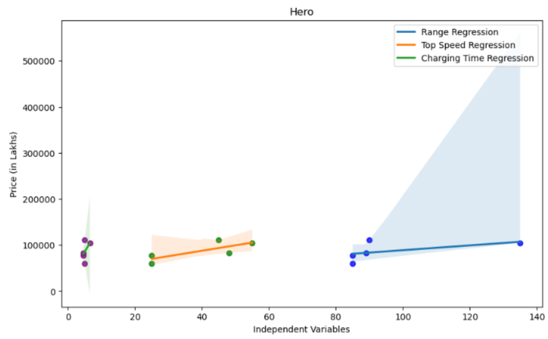

Fig. 4. Regression plot of Hero bike models

Fig. 3. The heatmap for Hero electric bikes reveals a strong positive correlation between range and charging time. There is also a moderate positive correlation between price and top speed.

Fig. 4. The plot illustrates the relationships between price and three independent variables (range, top speed, and charging time) for Hero electric bikes. The regression lines and confidence intervals suggest that price increases with range and charging time but decreases with top speed.

Ola :

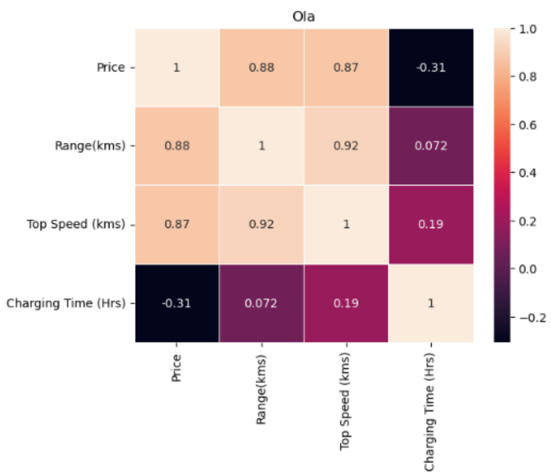

Fig. 5. Correlation plot of Ola bike models

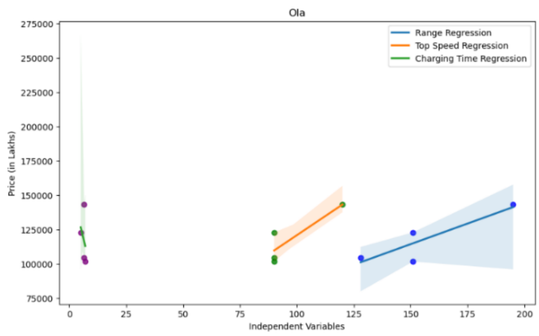

Fig. 6. Regression plot of Ola bike models

Fig. 5. The heatmap for Ola electric bikes reveals a strong positive correlation between range and top speed. There is also a moderate positive correlation between price and range. A moderate negative correlation is observed between price and charging time.

Fig. 6. The plot illustrates the relationships between price and three independent variables (range, top speed, and charging time) for Ola electric bikes. The regression lines and confidence intervals suggest that price increases with range and charging time but decreases with top speed.

Revolt :

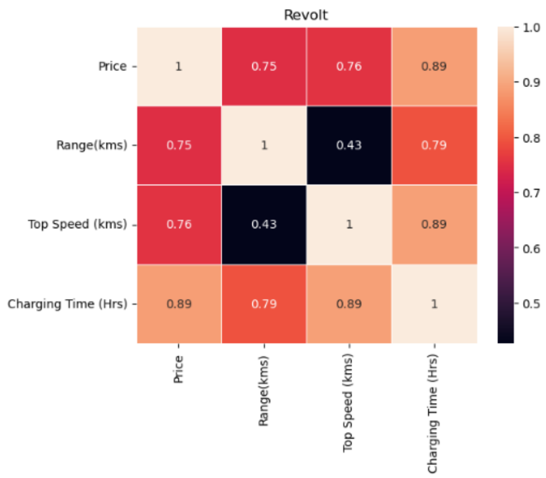

Fig. 7. Correlation plot of Revolt bike models

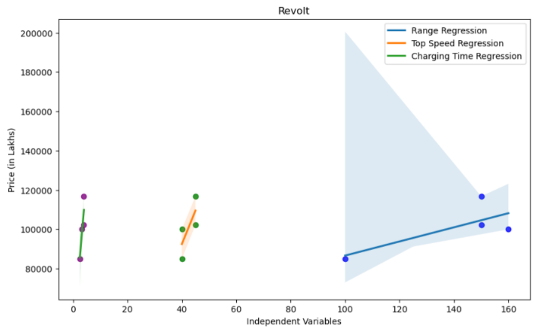

Fig. 8. Regression plot of Revolt bike models

Fig. 7. The heatmap for Revolt electric bikes reveals a strong positive correlation between price and charging time. There is also a moderate positive correlation between price and range. A moderate negative correlation is observed between price and top speed.

Fig. 8. The plot illustrates the relationships between price and three independent variables (range, top speed, and charging time) for Revolt electric bikes. The regression lines and confidence intervals suggest that price increases with range and charging time but decreases with top speed.

Overall Performance

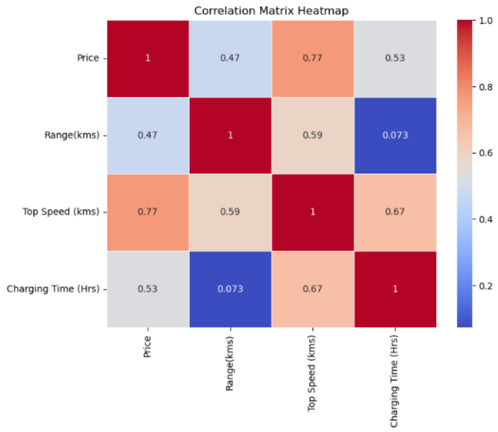

Fig. 9. Correlation plot of Overall bike models

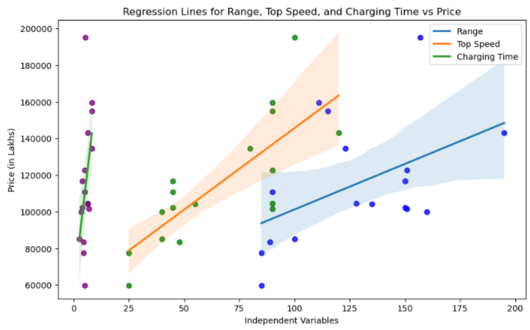

Fig. 10. Regression plot of Overall bike models

Fig. 9. The heatmap for overall electric bike models reveals a strong positive correlation between price and top speed. There is also a moderate positive correlation between price and range. A moderate negative correlation is observed between price and charging time.

Fig. 10. The plot illustrates the relationships between price and three independent variables (range, top speed, and charging time) for overall electric bike models. The regression lines and confidence intervals suggest that price increases with range and charging time but decreases with top speed.

5. Conclusion

Based on the analysis of the electric bike models, several key findings emerge that can guide consumers in choosing the best model suited to their needs. The Hero Electric Flash LX emerges as the best low-price model at ₹59,640, offering a modest range of 85 km and a top speed of 25 km/h, making it an affordable entry-level option for budget-conscious buyers. For those seeking best range, the Ola Electric S1 Pro stands out, with an impressive 195 km range, though it comes at a higher price of ₹1,43,098. This model is ideal for users who need long-distance travel capability.

In terms of highest top speed, the Ola Electric S1 Pro also leads, with a top speed of 120 km/h, making it suitable for users prioritizing speed. On the other hand, for fastest charging, the Revolt RV1 excels with a rapid charging time of just 2.5 hours, although it is priced at ₹84,990 and offers a range of 100 km.

Taking into account all the key criteria, the Ideal Model is proposed for users who want the best of all worlds: Hero Electric Flash LX's low price combined with the Ola Electric S1 Pro's range and speed, as well as Revolt RV1's fast charging. This ideal model would have a price of ₹59,640, a range of 195 km, a top speed of 120 km/h, and a charging time of 2.5

hours, making it the optimal choice for users seeking a balance of affordability, performance, and efficiency.

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